

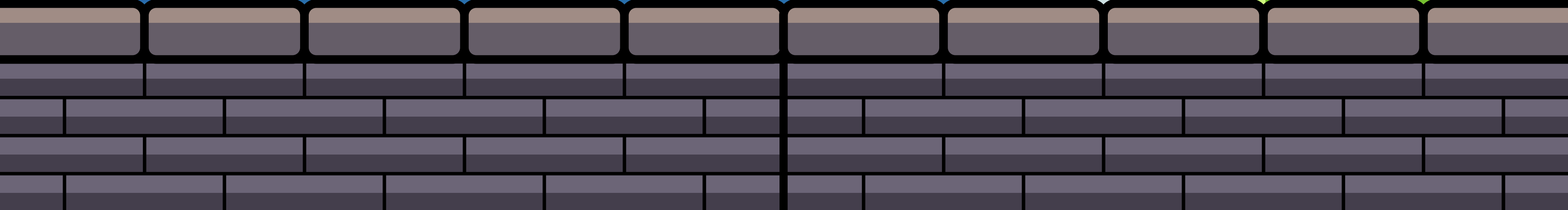
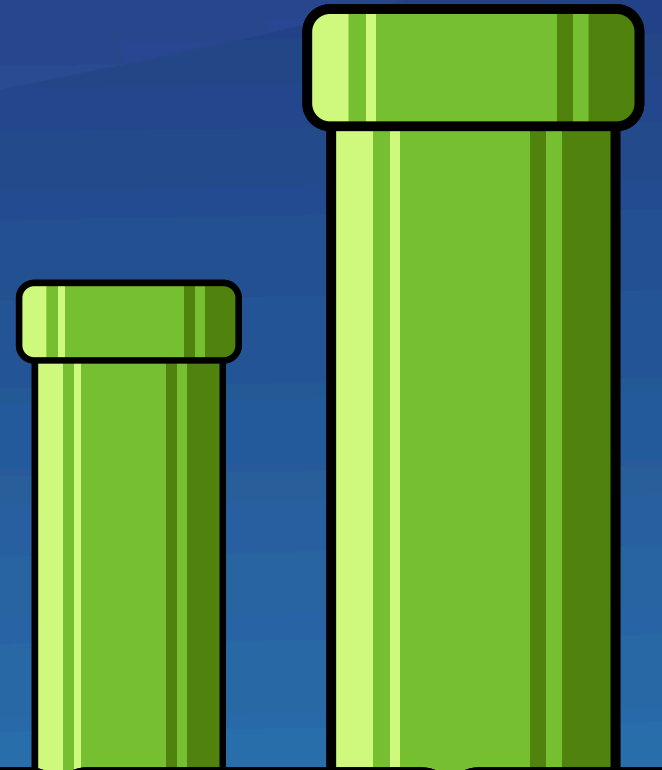
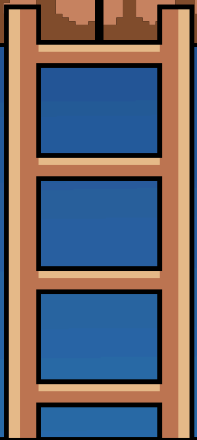
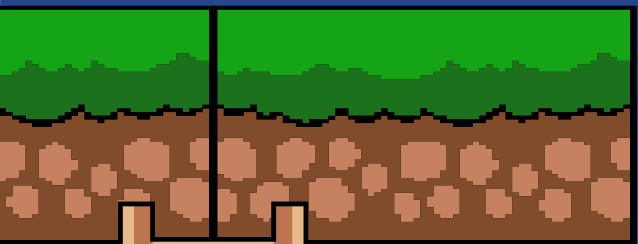
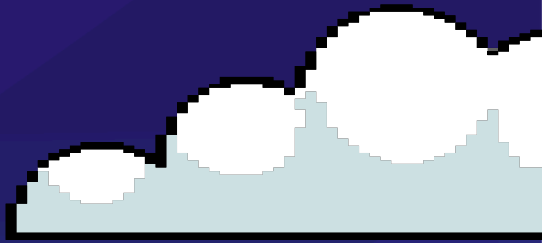
HDL PROJECT

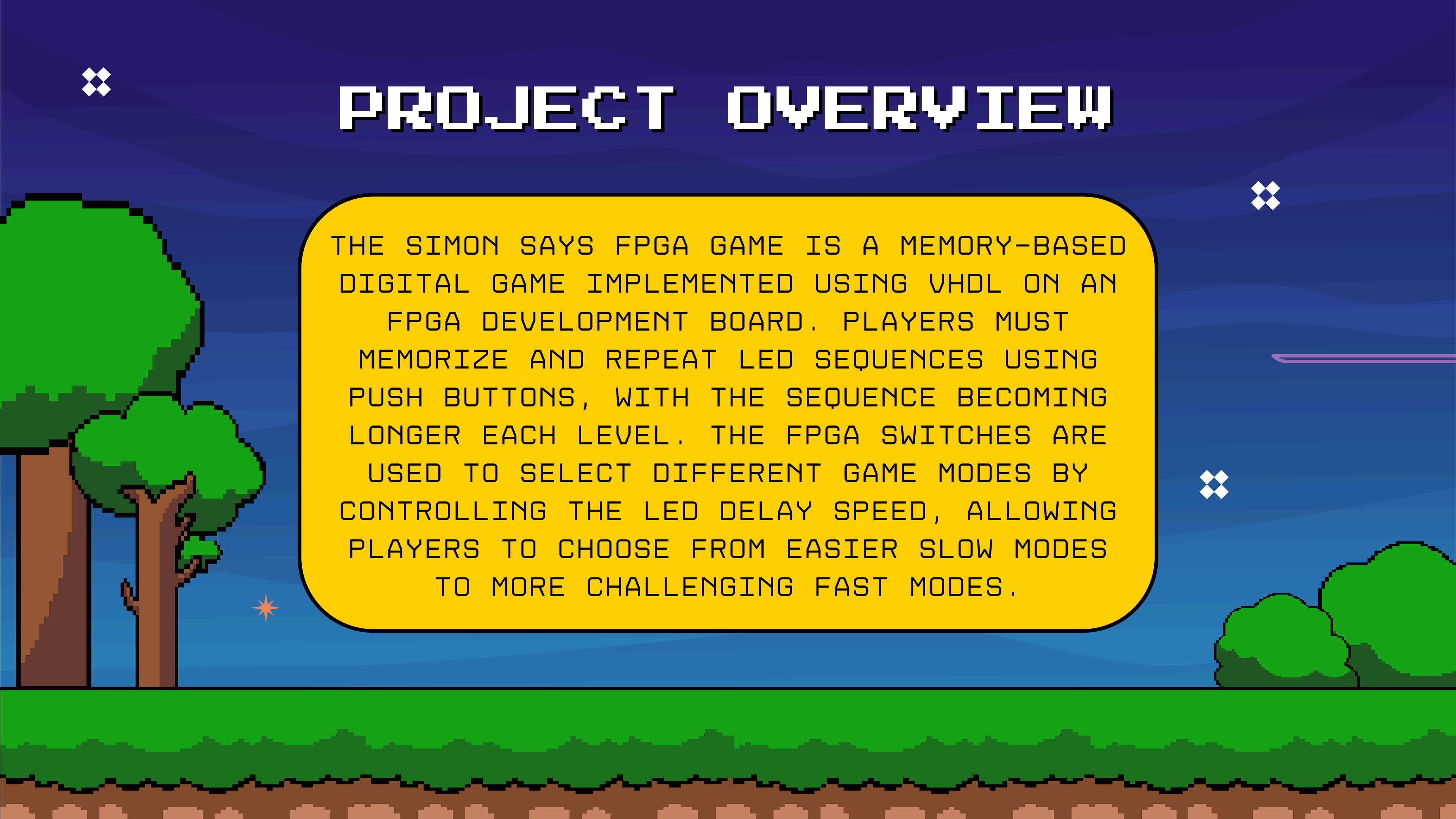
✦ Simon Says FPGA Game

✦ Memory Game



Are You
Ready To
Play?





PROJECT OVERVIEW

THE SIMON SAYS FPGA GAME IS A MEMORY-BASED DIGITAL GAME IMPLEMENTED USING VHDL ON AN FPGA DEVELOPMENT BOARD. PLAYERS MUST MEMORIZE AND REPEAT LED SEQUENCES USING PUSH BUTTONS, WITH THE SEQUENCE BECOMING LONGER EACH LEVEL. THE FPGA SWITCHES ARE USED TO SELECT DIFFERENT GAME MODES BY CONTROLLING THE LED DELAY SPEED, ALLOWING PLAYERS TO CHOOSE FROM EASIER SLOW MODES TO MORE CHALLENGING FAST MODES.

❖ OBJECTIVES

- ✓ DEVELOP A FUNCTIONAL SIMON SAYS GAME USING VHDL AND FPGA HARDWARE
- ✓ IMPLEMENT MEMORY SEQUENCE GENERATION AND USER VALIDATION
- ✓ PROVIDE MULTIPLE DIFFICULTY MODES USING FPGA SWITCHES
- ✓ DISPLAY REAL-TIME GAME INFORMATION
- ✓ DEMONSTRATE FPGA-BASED FINITE STATE MACHINE DESIGN



FEATURES

RANDOM
SEQUENCE
GENERATION

MULTIPLE
DIFFICULTY
MODES

LOCKED
DIFFICULTY
SYSTEM

LED SEQUENCE
DISPLAY

PUSH BUTTON
INPUT

FEATURES

LEVEL
PROGRESSION

GAME OVER
DETECTION

RESTART
FUNCTION

7-SEGMENT
DISPLAY
SUPPORT

FPGA BUILT IN FEATURES



LEDS (LEDR)

PUSH BUTTONS (KEY)

SWITCHES (SW)

7-SEGMENT DISPLAYS (HEX0, HEX1)

50 MHZ CLOCK (CLOCK_50)

INTERNAL FPGA FEATURES

FINITE STATE MACHINE (FSM)

LFSR RANDOM GENERATOR

COUNTERS/TIMERS

REGISTERS/MEMORY

The background is a pixel art illustration of a night scene. The sky is dark blue with horizontal bands of slightly different shades. There are white, pixelated clouds on the left and right. In the center, the words "Thank You" are written in a large, white, pixelated font with black outlines. Below the text is a yellow rounded rectangle containing the text "GREAT JOB! YOU HAVE TESTED YOUR MEMORY SKILLS." in a black, pixelated font. The foreground consists of a green grassy area with a brown, pixelated ground line at the bottom. On the left, there are two green trees with brown trunks. On the right, there are two purple buildings with white windows. Several small, white, four-pointed star-like symbols are scattered across the sky, along with a few small orange star-like symbols.

Thank You

GREAT JOB! YOU HAVE TESTED YOUR MEMORY SKILLS.